

Initial Project Planning Template

Date	04 July 2026
Team ID	ramanharshitha2005@gmail.com
Project Name	OptiCrop - Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

Sprint	Functional Req. (Epic)	User Story No.	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Environment Setup	USN-1	As an intern, I set up Python 3.10+, create a virtual environment, install all dependencies from requirements.txt, and verify the project runs at http://127.0.0.1:8000.	2	High	Raman Harshitha	01-07-26	01-07-26
Sprint-1	Dataset Prep & EDA	USN-2	As an intern, I download Crop_recommendation.csv from Kaggle, perform EDA (shape, dtypes, null check, class balance, correlation heatmap), and confirm 2200 rows, 22 balanced classes, no nulls.	3	High	Raman Harshitha	01-07-26	01-07-26
Sprint-1	Crop ML Training	USN-3	As an intern, I train 3 classifiers (RandomForest n=350, DecisionTree, SVC) on the 9-feature crop dataset, evaluate accuracy for each, save the ensemble bundle as crop_model.joblib, and generate confusion matrix PNG.	3	High	Raman Harshitha	02-07-26	02-07-26
Sprint-2	Disease ML Training	USN-4	As an intern, I augment the opticrop-data.json disease profiles (10x per record), encode features, train a Random Forest classifier (n=250), evaluate accuracy, and save disease_model.joblib.	3	High	Raman Harshitha	03-07-26	03-07-26
Sprint-2	Image Disease Model	USN-5	As a farmer, I want to upload a leaf photo and get the disease name so I can act immediately. (Train optional CNN or pixel-feature RF on PlantVillage dataset; save as image_disease_model.joblib or .h5.)	4	Medium	Raman Harshitha	03-07-26	03-07-26
Sprint-3	server.py — Core API	USN-6	As an intern, I implement all 7 REST API routes in server.py: /api/recommend, /api/suitability, /api/fertilizer, /api/disease, /api/image-disease, /api/weather, /api/analytics — with input validation and graceful error responses.	5	High	Raman Harshitha	04-07-26	04-07-26

Sprint-3	Weather Integration	USN-7	As a farmer, I want the form to auto-fill temperature, humidity, and rainfall from my city or GPS so I don't need to look them up manually. (Implement /api/weather using Open-Meteo geocoding + forecast API.)	2	High	Raman Harshitha	04-07-26	04-07-26
Sprint-4	Frontend — Recommend Page	USN-8	As a farmer, I want a web form where I enter soil and climate values and see the recommended crop, confidence score, and fertilizer advice instantly. (Build recommend.html with Scenario 1 and Scenario 2 tabs.)	3	High	Raman Harshitha	04-07-26	04-07-26
Sprint-4	Frontend — Disease Page	USN-9	As a farmer, I want to select my crop and symptoms (or upload a leaf photo) to detect the likely disease and see treatment suggestions. (Build disease.html with symptom form and image upload panel.)	3	High	Raman Harshitha	04-07-26	04-07-26
Sprint-5	Frontend — Analytics Dashboard	USN-10	As a researcher, I want to filter crop-environment data by crop, region, and season and see bar charts, radar charts, and scatter plots so I can identify patterns for policy planning. (Build insights.html with Chart.js and CSV export.)	4	High	Raman Harshitha	04-07-26	04-07-26
Sprint-5	Testing & Validation	USN-11	As an intern, I test all 7 API endpoints manually, verify correct responses for valid and invalid inputs, check error handling (missing model files, bad JSON, invalid image), and confirm the app runs without errors.	3	High	Raman Harshitha	04-07-26	04-07-26
Sprint-5	Documentation & GitHub	USN-12	As an intern, I write a complete README.md with setup instructions, push all code to GitHub (https://github.com/RamanHarshitha/Smart_Bridge), and fill in all SmartBridge project template documents.	2	High	Raman Harshitha	05-07-26	05-07-26